

Báo cáo thực hành backend 1

1. Code

- app.js

```
JS app.js X
src > JS app.js > [E] default
1  import express from "express";
2  import cors from "cors";
3
4  import contactsRouter from "../routes/contact.route.js"
5
6  import { notFound, errorHandler } from "../middlewares/error.middleware.js";
7
8  const app = express();
9  app.use(express.json());
10 app.use(cors());
11
12 // API
13 app.get("/", (request, response) => {
14   |   response.status(200).json({message: "Welcome to contact book application"})
15 })
16
17 app.use("/api/contacts", contactsRouter);
18
19 app.use(notFound);
20 app.use(errorHandler);
21
22 export default app;
```

- index.js

```
JS app.js JS index.js X
JS index.js > ...
1  import "dotenv/config"
2
3  import app from "../src/app.js"
4  import { connectDatabase } from "../src/config/database.js";
5
6  const PORT = process.env.PORT || 3000;
7  const URI = process.env.URI;
8
9  await connectDatabase(URI);
10 app.listen(PORT, () => {
11   |   console.log(`Server is running on port ${PORT}...`);
12 })
```

- .env.example

```
JS app.js .env.example X
.env.example
1 PORT=3000
2 URI=mongodb://localhost:27017/contactbook
```

- contact.controller.js

```
JS app.js JS contact.controller.js X
src > controllers > JS contact.controller.js > findAll
1 import * as contactService from "../services/contact.service.js"
2
3 import {ApiError} from "../middlewares/error.middleware.js"
4
5 export const create = async (request, response) => {
6   const payload = request.body
7   const responseContent = await contactService.create(payload)
8   response.status(201).json({data: responseContent});
9 }
10
11 export const findAll = async (request, response) => {
12   const contacts = await contactService.findAll();
13   response.status(200).json({data: contacts});
14 }
15
16 export const deleteAll = async (request, response) => {
17   await contactService.deleteAll();
18   response.status(204).send();
19 }
20
21 export const findAllFavorite = async (request, response) => {
22   const favoriteContacts = await contactService.findAllFavorite();
23   response.status(200).json({data: favoriteContacts});
24 }
25
26 export const findOne = async (request, response) => {
27   const id = request.params.id;
28   const contact = await contactService.findOne(id)
29   if (!contact) {
30     throw new ApiError(404, "Contact not found");
31   }
32   response.status(200).json({data: contact});
33 }
34
35 export const update = async (request, response) => {
36   const id = request.params.id;
37   const payload = request.body;
38   const contact = await contactService.update(id, payload);
39   if (!contact) {
40     throw new ApiError(404, "Contact not found");
41   }
42   response.status(200).json({data: contact});
43 }
44 }
```

```
JS app.js JS contact.controller.js X
src > controllers > JS contact.controller.js > findAll
25
26 export const findOne = async (request, response) => {
27   const id = request.params.id;
28   const contact = await contactService.findOne(id)
29   if (!contact) {
30     throw new ApiError(404, "Contact not found");
31   }
32   response.status(200).json({data: contact});
33 }
34
35 export const update = async (request, response) => {
36   const id = request.params.id;
37   const payload = request.body;
38   const contact = await contactService.update(id, payload);
39   if (!contact) {
40     throw new ApiError(404, "Contact not found");
41   }
42   response.status(200).json({data: contact});
43 }
44
45 export const deleteOne = async (request, response) => {
46   const id = request.params.id;
47   await contactService.deleteOne(id);
48   response.status(204).send();
49 }
50
51
```

- contact.route.js

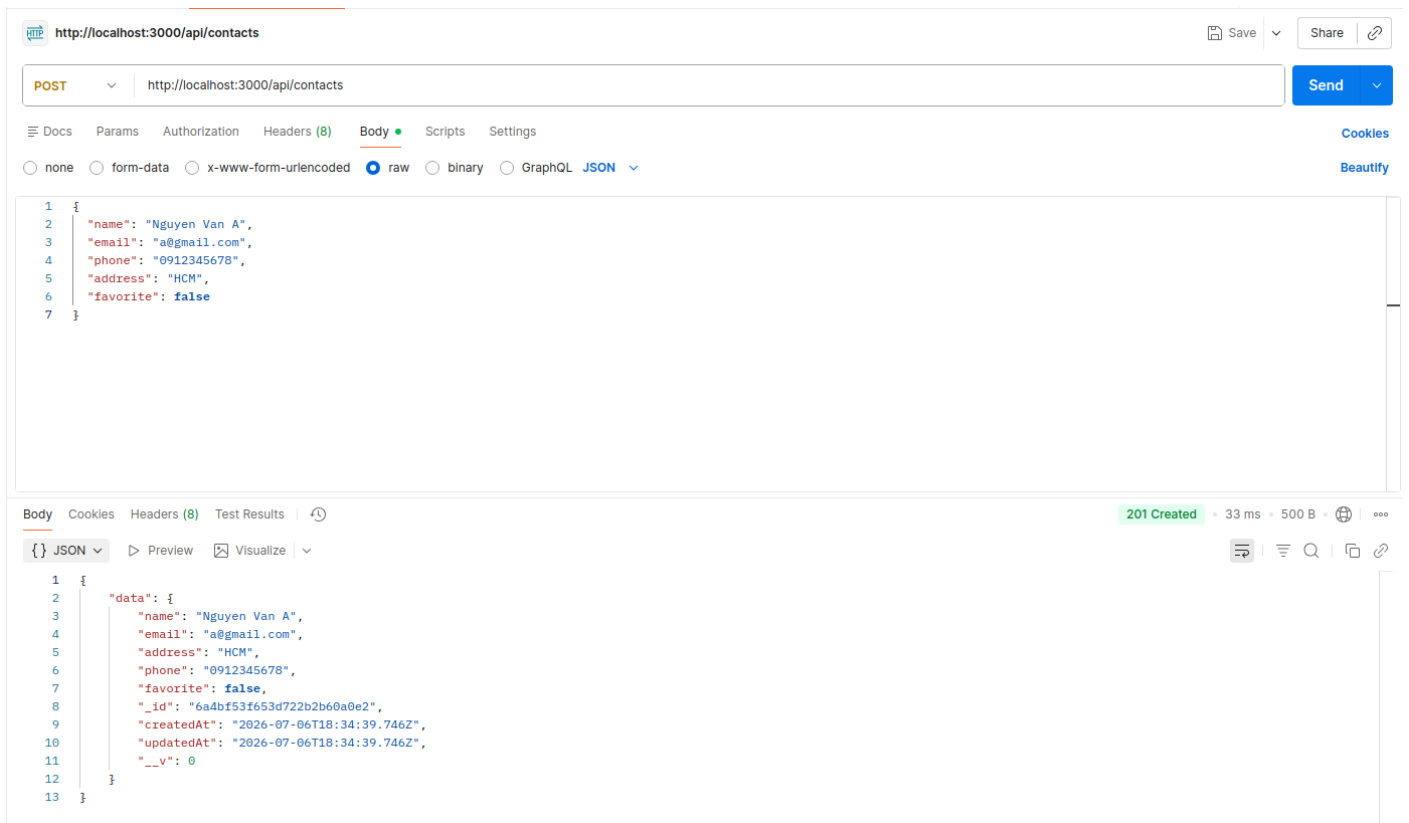
```
JS app.js JS contact.route.js X
src > routes > JS contact.route.js > default
1 import { Router } from "express";
2
3 import * as contacts from "../controllers/contact.controller.js"
4
5 const router = Router();
6
7 router.route("/")
8   .get(contacts.findAll)
9   .post(contacts.create)
10  .delete(contacts.deleteAll);
11
12 router.route("/favorite").get(contacts.findAllFavorite);
13
14 router.route("/:id")
15   .get(contacts.findOne)
16   .put(contacts.update)
17   .delete(contacts.deleteOne);
18
19 export default router;
```

- error.middleware.js

```
src > middlewares > JS error.middleware.js > errorHandler
1  export class ApiError extends Error {
2    constructor(statusCode, message) {
3      super(message);
4      this.statusCode = statusCode;
5      this.name = "ApiError";
6    }
7  }
8
9  export const notFound = (request, response, next) => {
10   next(new ApiError(404, "Resource not found"));
11 }
12
13 export const errorHandler = (error, request, response, next) => {
14   const statusCode = error.statusCode ?? 500;
15   const message = error.message ?? "Internal Server Error";
16
17   if (statusCode >= 500) {
18     console.error(error);
19   }
20
21   response.status(statusCode).json({error: {message}});
22 }
```

2. Test trên Postman

- POST <http://localhost:3000/api/contacts>



- GET <http://localhost:3000/api/contacts>

http://localhost:3000/api/contacts

GET http://localhost:3000/api/contacts

Docs Params Authorization Headers (6) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

1 Ctrl+Alt+P to Ask AI

Body Cookies Headers (8) Test Results

200 OK • 9 ms • 497 B

JSON Preview Visualize

```
1 {
2   "data": [
3     {
4       "_id": "6a4bf53f653d722b2b60a0e2",
5       "name": "Nguyen Van A",
6       "email": "a@gmail.com",
7       "address": "HCM",
8       "phone": "0912345678",
9       "favorite": false,
10      "createdAt": "2026-07-06T18:34:39.746Z",
11      "updatedAt": "2026-07-06T18:34:39.746Z",
12      "__v": 0
13    }
14  ]
15 }
```

- GET <http://localhost:3000/api/contacts/favorite>

http://localhost:3000/api/contacts/favorite

GET http://localhost:3000/api/contacts/favorite

Docs Params Authorization Headers (6) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

1 Ctrl+Alt+P to Ask AI

Body Cookies Headers (8) Test Results

200 OK • 9 ms • 277 B

JSON Preview Visualize

```
1 {
2   "data": []
3 }
```

- GET <http://localhost:3000/api/contacts/<contact-id>>

http://localhost:3000/api/contacts/6a4bf53f653d722b2b60a0e2

GET http://localhost:3000/api/contacts/6a4bf53f653d722b2b60a0e2

Body

1 Ctrl+Alt+P to Ask AI

Body Cookies Headers (8) Test Results

200 OK 7 ms 495 B

```
1 {
2   "data": {
3     "_id": "6a4bf53f653d722b2b60a0e2",
4     "name": "Nguyen Van A",
5     "email": "a@gmail.com",
6     "address": "HCM",
7     "phone": "0912345678",
8     "favorite": false,
9     "createdAt": "2026-07-06T18:34:39.746Z",
10    "updatedAt": "2026-07-06T18:34:39.746Z",
11    "__v": 0
12  }
13 }
```

- PUT <http://localhost:3000/api/contacts/<contact-id>>

http://localhost:3000/api/contacts/6a4bf53f653d722b2b60a0e2

PUT http://localhost:3000/api/contacts/6a4bf53f653d722b2b60a0e2

Body

```
1 {
2   "name": "Nguyen Van A Updated",
3   "email": "a@gmail.com",
4   "phone": "0987654321",
5   "address": "Da Nang",
6   "favorite": true
7 }
```

Body Cookies Headers (8) Test Results

200 OK 8 ms 506 B

```
1 {
2   "data": {
3     "_id": "6a4bf53f653d722b2b60a0e2",
4     "name": "Nguyen Van A Updated",
5     "email": "a@gmail.com",
6     "address": "Da Nang",
7     "phone": "0987654321",
8     "favorite": true,
9     "createdAt": "2026-07-06T18:34:39.746Z",
10    "updatedAt": "2026-07-06T18:39:56.306Z",
11    "__v": 0
12  }
13 }
```

- DELETE <http://localhost:3000/api/contacts/<contact-id>>

http://localhost:3000/api/contacts/6a4bf53f653d722b2b60a0e2

DELETE http://localhost:3000/api/contacts/6a4bf53f653d722b2b60a0e2

Docs Params Authorization Headers (8) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 {
2   "name": "Nguyen Van A Updated",
3   "email": "a@gmail.com",
4   "phone": "0987654321",
5   "address": "Da Nang",
6   "favorite": true
7 }
```

Body Cookies Headers (5) Test Results

204 No Content 7 ms 166 B

Raw Preview Visualize

1

- DELETE <http://localhost:3000/api/contacts>

http://localhost:3000/api/contacts

DELETE http://localhost:3000/api/contacts

Docs Params Authorization Headers (8) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 {
2   "name": "Nguyen Van A Updated",
3   "email": "a@gmail.com",
4   "phone": "0987654321",
5   "address": "Da Nang",
6   "favorite": true
7 }
```

Body Cookies Headers (5) Test Results

204 No Content 8 ms 166 B

Raw Preview Visualize

1

- GET <http://localhost:3000/huhukheckhec/kakaka>

 http://localhost:3000/huhukheckhec/kakaka

Save

Share

DELETE

http://localhost:3000/huhukheckhec/kakaka

Send

DocsParamsAuthorizationHeaders (8)BodyScriptsSettings

noneform-datax-www-form-urlencodedrawbinaryGraphQLJSON

Beautify

```
1 {
2   "name": "Nguyen Van A Updated",
3   "email": "a@gmail.com",
4   "phone": "0987654321",
5   "address": "Da Nang",
6   "favorite": true
7 }
```

BodyCookiesHeaders (8)Test Results

404 Not Found • 5 ms • 316 B

{ }JSON

Preview

Debug with AI

1 {
2 "error": {
3 "message": "Resource not found"
4 }
5 }